Vivify Admission Repair and Tier-D Rank-24 Closure

Tactix engineering handoff

20 August 2026

Objective and outcome

The objective was to start the next immutable Tier-D closure at rank 24, compare Tactix and Kissat from the latest common semantic state, and repair only the earliest portable scheduling, representation, or semantic divergence. The investigation specifically audited Vivify admission, cadence, candidate ordering, tier shares, and its placement among adjacent proof rules.

The accepted repair removes every formula-selected Vivify disable. It is a general source-parity correction: formula classification may dose BVE, Factor, forward work, and Vivify effort, but cannot remove the Vivify opportunity that Kissat offers in every search-time Probe phase. The repair closes rank 24 UNSAT in 356.205 seconds with an independently verified proof. The cumulative ranks-21–40 gate preserves all six inherited closures and also closes rank 34 SAT and rank 39 UNSAT. The accepted set is now ranks 21, 23, 24, 25, 28, 29, 34, 38, and 39; eleven ranks remain open.

Root cause and design decision

The immutable rank-24 formula contains 167,755 variables and 473,169 clauses. The old formula classifier placed every formula in the 120,000–199,999 variable range into a `NoVivify` profile. During scheduler construction, that profile omitted the Vivify entry from all search-time Probe phases.

Kissat has no equivalent variable-range or density-range admission policy. Its `probe.c` calls `kissat_vivify` unconditionally between the early Backbone and Sweep calls in every search-time Probe. The `vivify` option, 100-per-mille effort, 3/3/1 redundant-tier shares, irredundant share, and irredundant productivity delay bound the work inside that opportunity. Kissat's reduced startup Probe intentionally omits Vivify. Thus the earliest portable divergence was policy admission at the first due search-time Probe, not a Vivify rule-side guard.

The classifier's former `*NoVivify` variants now collapse onto the same profiles with only their independently supported BVE, Factor, forward, and dose dimensions retained. No benchmark name, hash, rank, deadline, literal, conflict, per-instance arm, or new tuned constant appears in the solver. Explicit typed configuration and `TACTIX_SLOT_DISABLE=Vivify` remain available as diagnostic and N1 selection surfaces.

Vivify parity audit

The repair is admission-only because the surrounding machinery is already source-aligned and regression-pinned. The exact default orders are:

Startup Probe. Congruence, Substitute, Backbone, Sweep, Substitute, Factor.

Search-time Probe. Congruence, Substitute, Backbone, Vivify, Sweep, Substitute, TransitiveReduce, Backbone, Factor.

Within Vivify, Tactix uses the focused dynamic tier limits; 3/3/1 redundant shares followed by the irredundant share; Kissat's leftover priority and whole-scope re-arm; occurrence-ranked candidate and literal order with the descending raw-literal tie; committed clause pose and rebuilt large watches; cross-candidate trail-prefix reuse; earliest implied-literal selection; deduction precedence; and the one-percent irredundant productivity delay. The stale source comment suggesting that `better_implied` was absent was corrected: the later trail-order scan already implements it.

`test_scheduling` now pins both complete default sequences. `test_strategy_config` sweeps every formula-profile boundary and requires default Vivify admission, while separately proving that a caller's explicit disable remains intact. Existing focused Vivify order, deduction, tier, N1, and proof tests remain unchanged.

Diagnostic Slurm job `1636836` ran the frozen rank-24 input through instrumented Kissat and the preceding Tactix source with symmetric trace controls. Both diagnostic tasks timed out at 1,200 seconds and left no certificate residue. The trace identifies Kissat's search-time Vivify boundary; the source classifier identifies why Tactix could not take it. No frozen reference performance result was rerun.

Correctness and iteration guard

Release passes 72/72 tests, including native LRAT and DRAT micro-proof replay. No fuzz or unit test exposed a soundness defect or crash, so the development plan's mandatory Claude repair delegation was not triggered.

Matched Slurm job `1636852` ran current shipped and candidate binaries in the same window under the pre-Phase-4.5 contract: two CPUs, 16 GiB, 1,200-second solver timeout, no node constraint, and no array throttle. It covered all 28 Tier-B fixtures and the predeclared fixed five mechanism-relevant Tier-C fixtures.

Tier	shipped checked	candidate checked	shipped wall	candidate wall
B	28/28	28/28	1228.741 s	649.578 s
C	5/5	5/5	1790.271 s	1586.259 s
Total	33/33	33/33	3019.012 s	2235.837 s

There is no solve loss. The largest improvement is `mod4block`, from 718.396 seconds to 95.820 seconds. The largest slowdown is `SC25_Timetable`, from 18.908 seconds to 67.615 seconds; it remains a checked SAT solve. The hard Tier-C `T105.2.0` UNSAT case is neutral at 807.998 versus 811.605 seconds. Every SAT model and UNSAT proof verifies, and all 66 tasks delete their scratch and certificate artifacts.

The immutable manifest is `benchmarks/runs/20260820-2306-BC-vivify-admission/manifest.json` (SHA-256 `f230612e...6216`). The candidate binary is `b6c991e0...fb6f`, from source commit `b7df39f8...b6985` plus source diff `180bf8a8...9f1`.

Rank-24 and cumulative Tier-D evidence

Targeted job `1636927` proves rank 24 UNSAT in 356.205 seconds. Vivify runs 208 times, checks 702,223 candidates, and performs 177,032 vivifications. `drat-trim` verifies the 1,398,139,701-byte

proof in 94.099 seconds: 1,746,149 of 8,524,960 lemmas enter the core, with 31,608,568 resolution steps and zero RAT core lemmas. The runner then deletes the proof and scratch directory. Its manifest is `benchmarks/runs/20260820-2322-tierd-rank24-vivify-admission/manifest.json` (`a7bc9530...e6be`).

Cumulative job `1636928` runs all frozen ranks 21–40. Nine answers are independently checked: ranks 21, 23, 24, 25, 28, 29, 34, 38, and 39. The six predecessors are preserved; ranks 24 and 39 are UNSAT gains and rank 34 is an incidental SAT gain. The other eleven tasks are clean 1,200-second UNKNOWN timeouts. All 20 tasks complete without infrastructure failure or certificate residue. Its manifest is `benchmarks/runs/20260820-2331-tierd-ranks21-40-vivify-admission/manifest.json` (`65e2c85f...550`).

The full frozen narrative and reopening condition are in `benchmarks/reports/2026-08-20-tierd-rank24-vivify-admission.md`. Durable policy ownership is recorded in `docs/implementation-notes.md`; T4 and the accepted N4 evidence pin are updated in `docs/scheduling-parity-ledger.md`.

Limitations, risks, and next actions

This repair does not claim exact runtime parity with Kissat, and it does not erase formula-profile dosing for independently measured mechanisms. One Tier-B instance slows substantially, although aggregate B and C time improves and coverage is unchanged. Candidate-order and deduction parity remain subject to their existing ledger reopening conditions.

Rank 22 remains parked at its previously localized common-input Vivify boundary; this admission repair does not by itself close that much larger instance. Following the requested UNSAT-first order, the next open targets are ranks 27, 31, 32, 33, 35, 37, and 40. Each future change still owes the full Tier-B plus predeclared mechanism-relevant Tier-C guard before its Tier-D witness. Reopen this repair if any formula classifier again clears `vivify_enabled`, if either default Probe sequence changes, if a checked predecessor is lost, or if proof/model validation or cleanup fails.